Week 3 build card

A removable cone pusher

Team ____ Builders A ____ B ____ C ____

Build an attachment that pushes a small cone into a target zone while keeping it upright. The cone slides on the floor; do not lift or carry it.

Materials: Beams, pins, braces or pre-cut card, craft tape, a mounting proxy, ruler, small cone and marked target. No extra motor or sensor.

Build and test

1. Make a low front bar and two secure mounting points. Use the proxy while the other team has the robot.

2. Hand-push the cone 10 cm toward the zone. Keep the bar off the floor and clear of the wheel outline. Start gently.

3. Repeat three times from the same start. Count success only when the cone stays upright and its whole base finishes inside the zone.

4. Change ONE feature: contact height, width, bracing, or a shallow V with room for the cone. Try three more times. Keep the cone, floor, start and hand-push distance the same.

Our jobs

Builder connects pieces. Designer selects parts and checks fit. Tester resets the cone and reports results. Rotate when the mentor calls time.

Show your evidence

First design: ____ / 3 successes. Changed design: ____ / 3.

We noticed _____. We changed _____.

Try it on the robot

Fit only while stopped, after a mentor checks wheels, cables, buttons and floor clearance. Use the straight cone recipe: repeat drive(60), pause(300) three times; then signal D once. Use await and the full supplied project. No turns. The mentor sets the target for the rehearsed travel.

Stop if the cone tips or escapes. Wait for motion to stop before resetting. Remove the pusher before running the dance. Finish by minute 50.